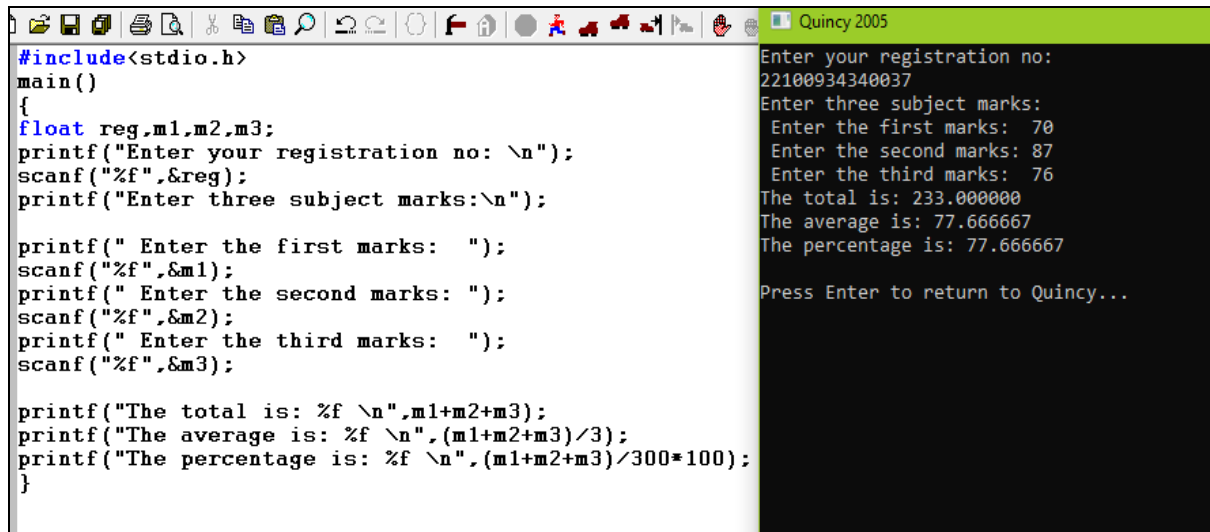


MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION SYSTEMS.

Programming Concepts

Discussed questions and their outputs.

1. Write a C program to read student registration number and marks of three subjects and calculate the total, percentage and division.



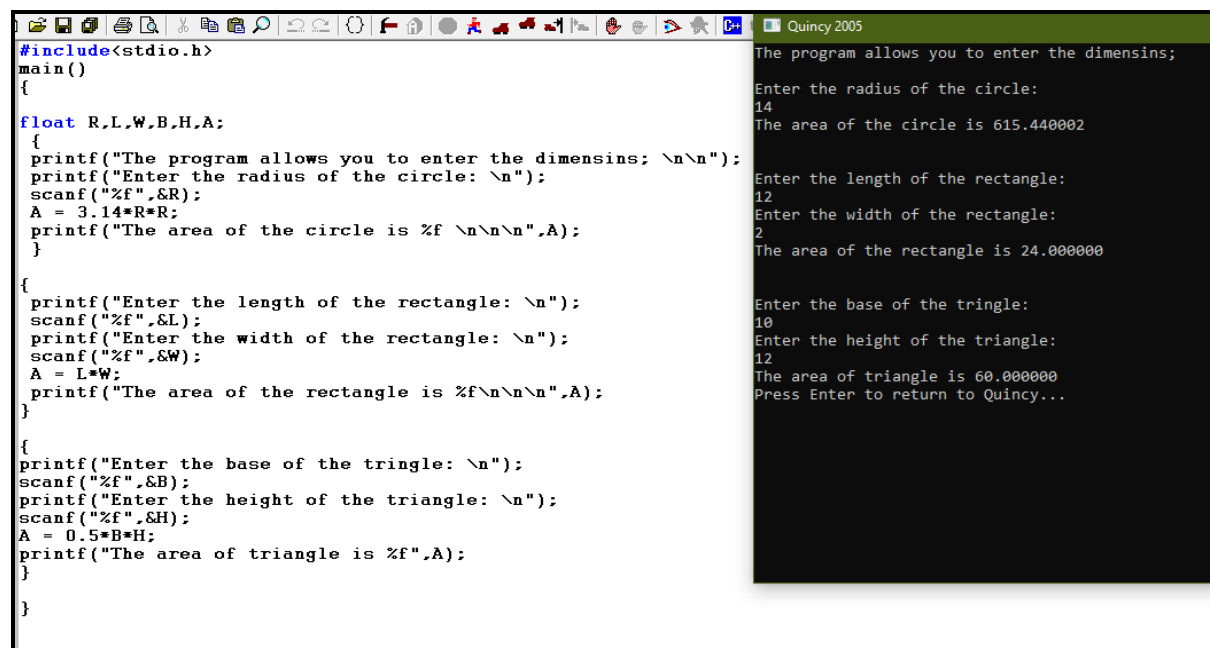
```
#include<stdio.h>
main()
{
    float reg,m1,m2,m3;
    printf("Enter your registration no: \n");
    scanf("%f",&reg);
    printf("Enter three subject marks:\n");

    printf(" Enter the first marks: ");
    scanf("%f",&m1);
    printf(" Enter the second marks: ");
    scanf("%f",&m2);
    printf(" Enter the third marks: ");
    scanf("%f",&m3);

    printf("The total is: %f \n",m1+m2+m3);
    printf("The average is: %f \n", (m1+m2+m3)/3);
    printf("The percentage is: %f \n", (m1+m2+m3)/300*100);
}
```

Enter your registration no:
22100934340037
Enter three subject marks:
Enter the first marks: 70
Enter the second marks: 87
Enter the third marks: 76
The total is: 233.000000
The average is: 77.666667
The percentage is: 77.666667
Press Enter to return to Quincey...

2. Write a program that computes the area of a circle, the area of a rectangle and the area of a triangle. Let the program prompt the user to enter dimensions, perform calculations and print the result on each case.



```
#include<stdio.h>
main()
{
    float R,L,W,B,H,A;
    {
        printf("The program allows you to enter the dimensins; \n\n");
        printf("Enter the radius of the circle: \n");
        scanf("%f",&R);
        A = 3.14*R*R;
        printf("The area of the circle is %f \n\n\n",A);
    }

    {
        printf("Enter the length of the rectangle: \n");
        scanf("%f",&L);
        printf("Enter the width of the rectangle: \n");
        scanf("%f",&W);
        A = L*W;
        printf("The area of the rectangle is %f\n\n\n",A);
    }

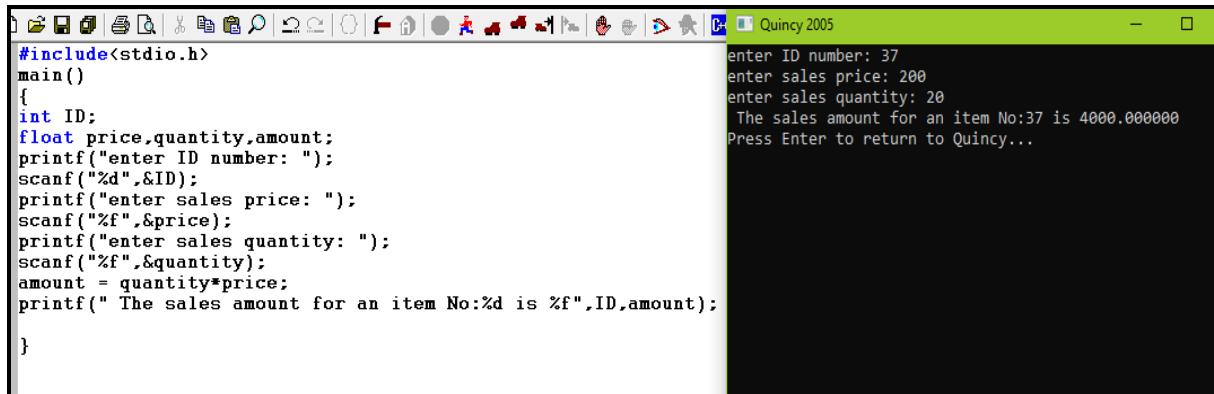
    {
        printf("Enter the base of the tringle: \n");
        scanf("%f",&B);
        printf("Enter the height of the triangle: \n");
        scanf("%f",&H);
        A = 0.5*B*H;
        printf("The area of triangle is %f",A);
    }
}
```

The program allows you to enter the dimensins;
Enter the radius of the circle:
14
The area of the circle is 615.440002
Enter the length of the rectangle:
12
Enter the width of the rectangle:
2
The area of the rectangle is 24.000000
Enter the base of the tringle:
10
Enter the height of the triangle:
12
The area of triangle is 60.000000
Press Enter to return to Quincey...

Light.

3. Write a C program that reads ID number, sales price and sales quantity of an item and compute sales amount. The display results should be in the following format;

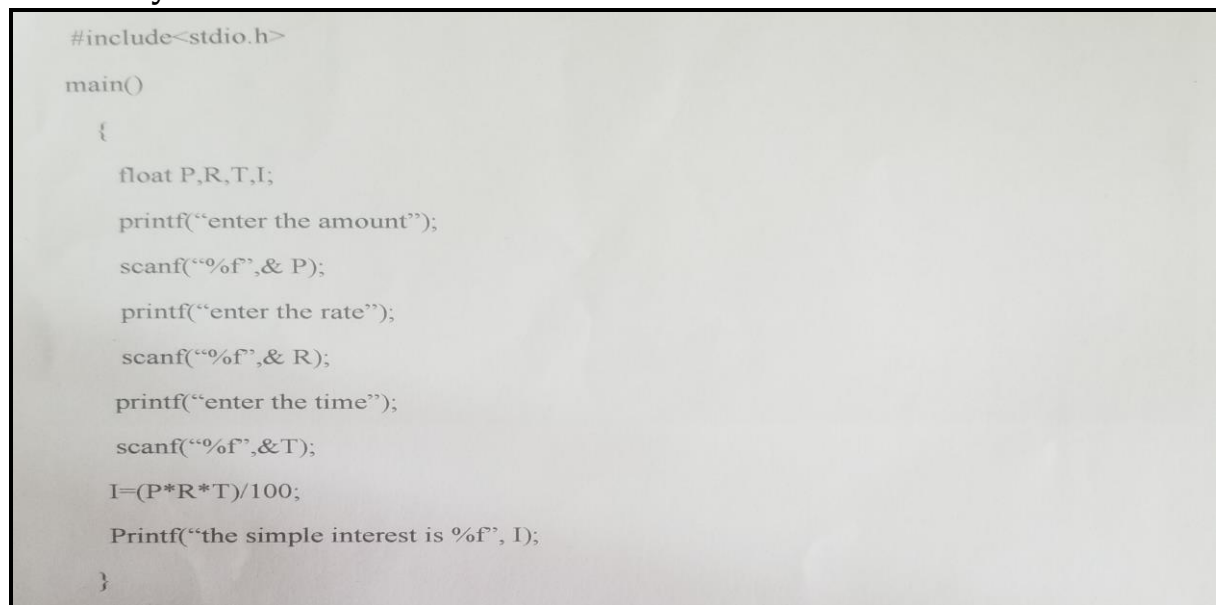
The sales amount for an Item No. xxxx is yyyy



```
#include<stdio.h>
main()
{
    int ID;
    float price,quantity,amount;
    printf("enter ID number: ");
    scanf("%d",&ID);
    printf("enter sales price: ");
    scanf("%f",&price);
    printf("enter sales quantity: ");
    scanf("%f",&quantity);
    amount = quantity*price;
    printf(" The sales amount for an item No:%d is %f",ID,amount);
}
```

enter ID number: 37
enter sales price: 200
enter sales quantity: 20
The sales amount for an item No:37 is 4000.000000
Press Enter to return to QuincY...

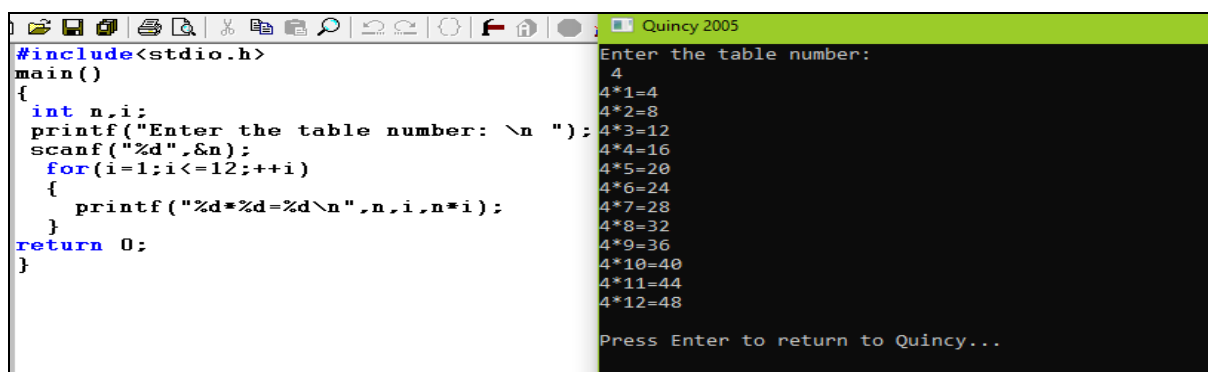
4. Write a C program to calculate simple interest of a given amount for a year entered by user.



```
#include<stdio.h>

main()
{
    float P,R,T,I;
    printf("enter the amount");
    scanf("%f",&P);
    printf("enter the rate");
    scanf("%f",&R);
    printf("enter the time");
    scanf("%f",&T);
    I=(P*R*T)/100;
    Printf("the simple interest is %f", I);
}
```

5. Write a C program to print multiplication table of any number.



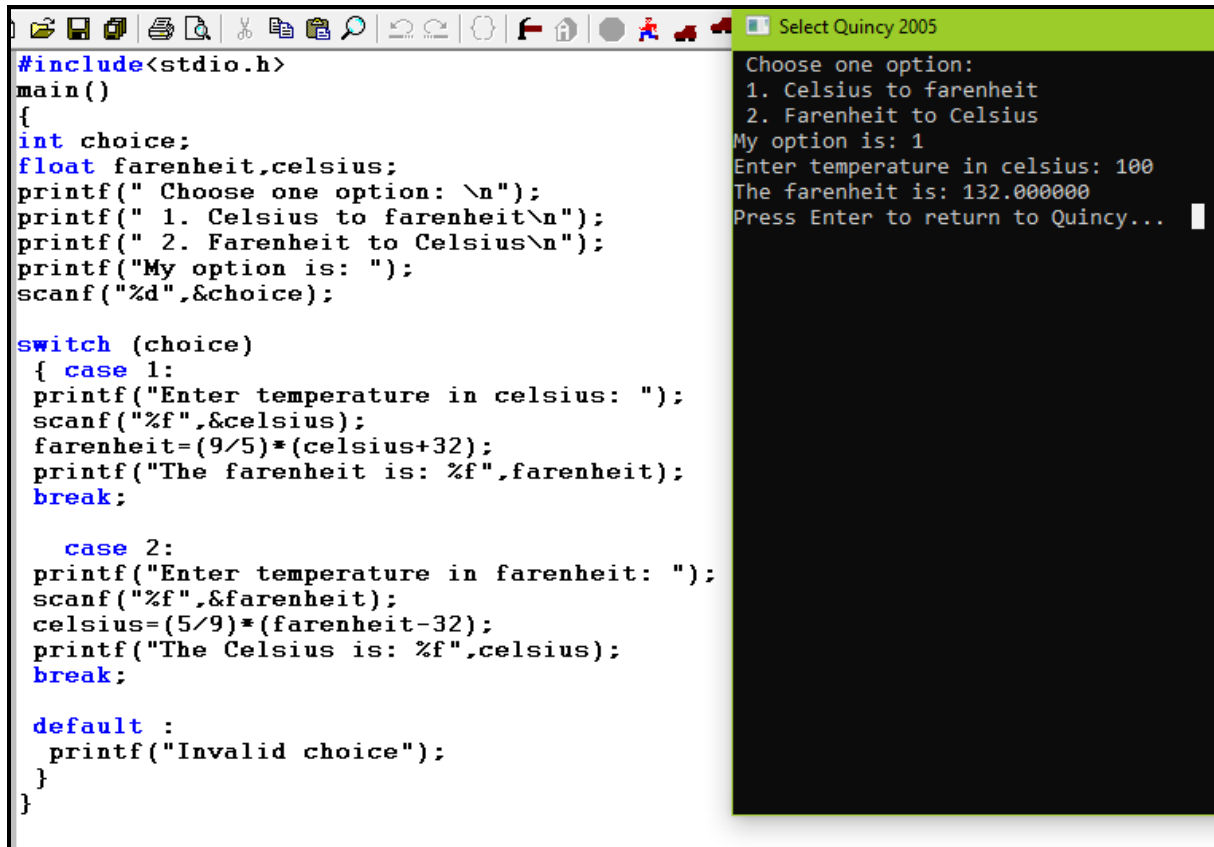
```
#include<stdio.h>
main()
{
    int n,i;
    printf("Enter the table number: \n ");
    scanf("%d",&n);
    for(i=1;i<=12;++i)
    {
        printf("%d*%d=%d\n",n,i,n*i);
    }
    return 0;
}
```

Enter the table number:
4
4*1=4
4*2=8
4*3=12
4*4=16
4*5=20
4*6=24
4*7=28
4*8=32
4*9=36
4*10=40
4*11=44
4*12=48
Press Enter to return to QuincY...

Light.

Light.

6. Using switch condition statement, write a C program to convert degrees Fahrenheit to degrees Centigrade or degrees Centigrade to degrees Fahrenheit. The program should display two menu options on which a user can make a choice on what conversion he/she want to make. Hint: $C = (5/9)(F - 32)$

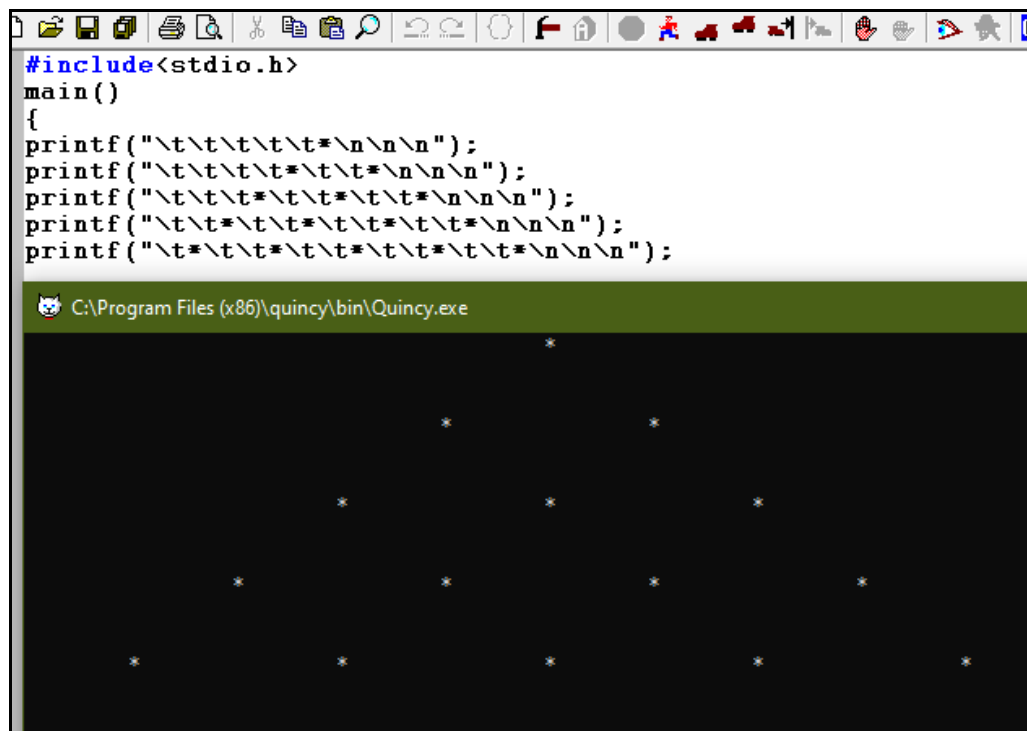
The image shows a screenshot of a code editor window titled "Select Quincy 2005". The left pane displays the C source code for a temperature conversion program. The code includes `<stdio.h>` and defines a `main()` function. It declares `int choice;` and `float fahrenheit, celsius;`. The program prompts the user to "Choose one option:" and lists two choices: "1. Celsius to fahrenheit" and "2. Fahrenheit to Celsius". It then reads the user's choice into `choice` using `scanf("%d", &choice);`. A `switch` statement follows, with `case 1:` for Celsius to Fahrenheit conversion and `case 2:` for Fahrenheit to Celsius conversion. Both cases use `printf` to prompt for input, `scanf` to read the value, and the appropriate conversion formula. The `default` case prints "Invalid choice". The right pane shows the program's execution output, which matches the code's logic: it prompts for an option, the user enters '1', it prompts for Celsius temperature (100), and it outputs the Fahrenheit result (132.000000).

7. a. Write a C program to display the following output;

```
*
* *
* * *
* * * *
* * * * *
```

Light.

Light.



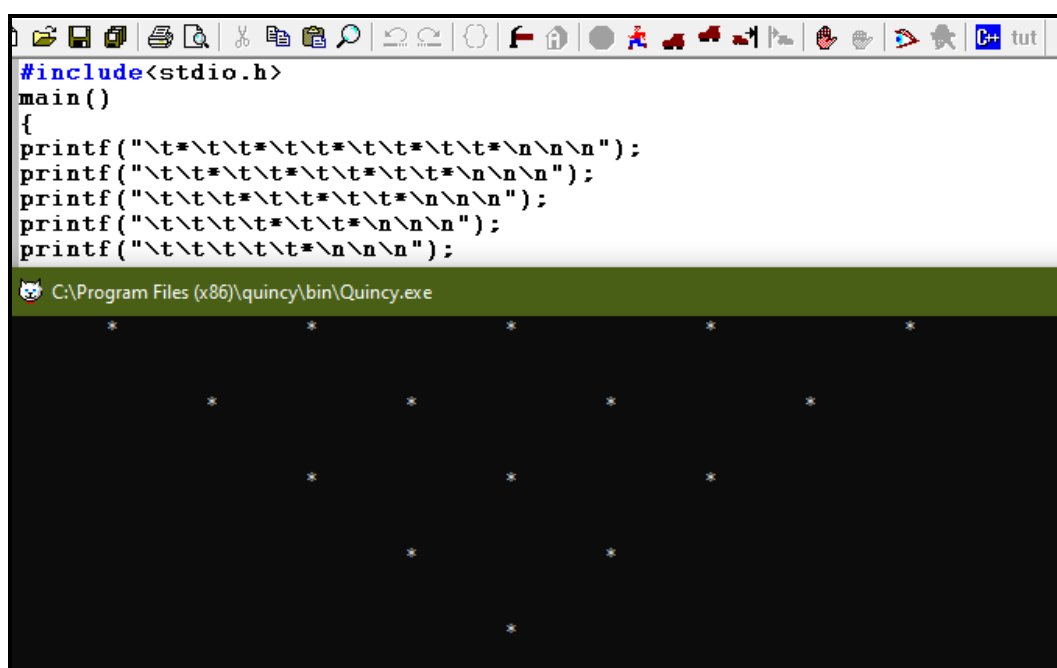
```
#include<stdio.h>
main()
{
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
}
```

C:\Program Files (x86)\quincy\bin\Quincy.exe

The output shows a 5x5 grid of asterisks on a black background, with each asterisk preceded by four tabs.

b. Write a C program to display the following output;

```
* * * * *
* * * *
* * *
* *
*
```



```
#include<stdio.h>
main()
{
printf("\t*\t*\t*\t*\t*\n\n\n");
printf("\t*\t*\t*\t*\t*\n\n\n");
printf("\t*\t*\t*\t*\t*\n\n\n");
printf("\t*\t*\t*\t*\t*\n\n\n");
printf("\t*\t*\t*\t*\t*\n\n\n");
}
```

C:\Program Files (x86)\quincy\bin\Quincy.exe

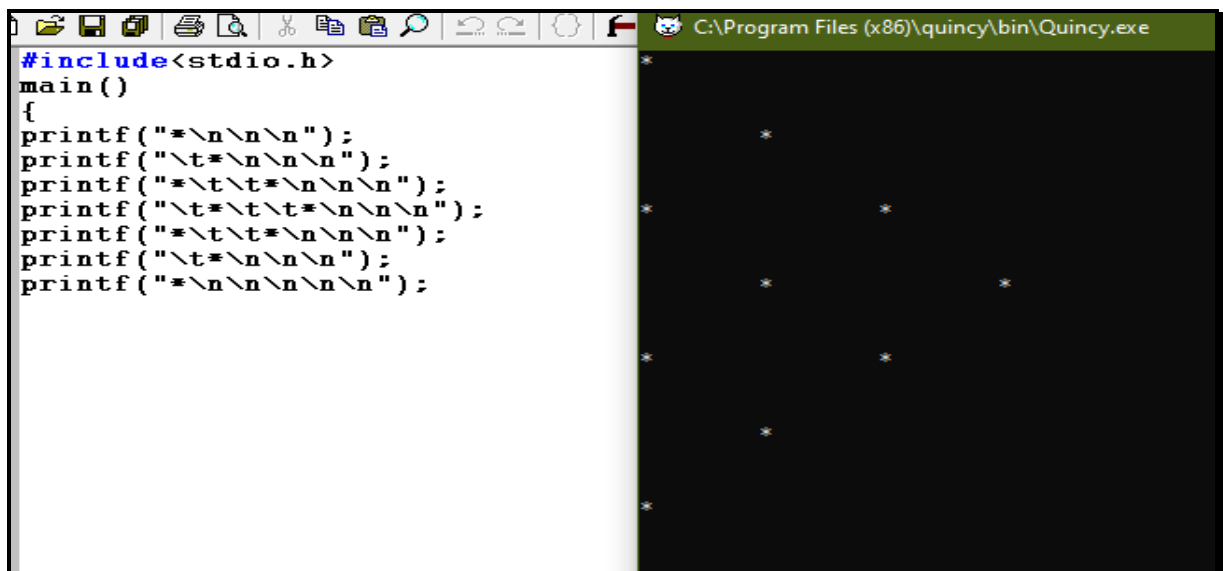
The output shows a 5x5 grid of asterisks on a black background, with each asterisk preceded by a tab.

Light.

Light.

c. Write a C program to display the following output;

```
*  
  
  *  
  
*      *  
  
  *      *  
  
*      *  
  
  *  
  
*
```



The screenshot shows a C program in a text editor and its output in a separate window. The program uses `printf` with escape sequences like `\n` and `\t` to create a pattern of asterisks. The output window shows the resulting pattern of asterisks on a black background.

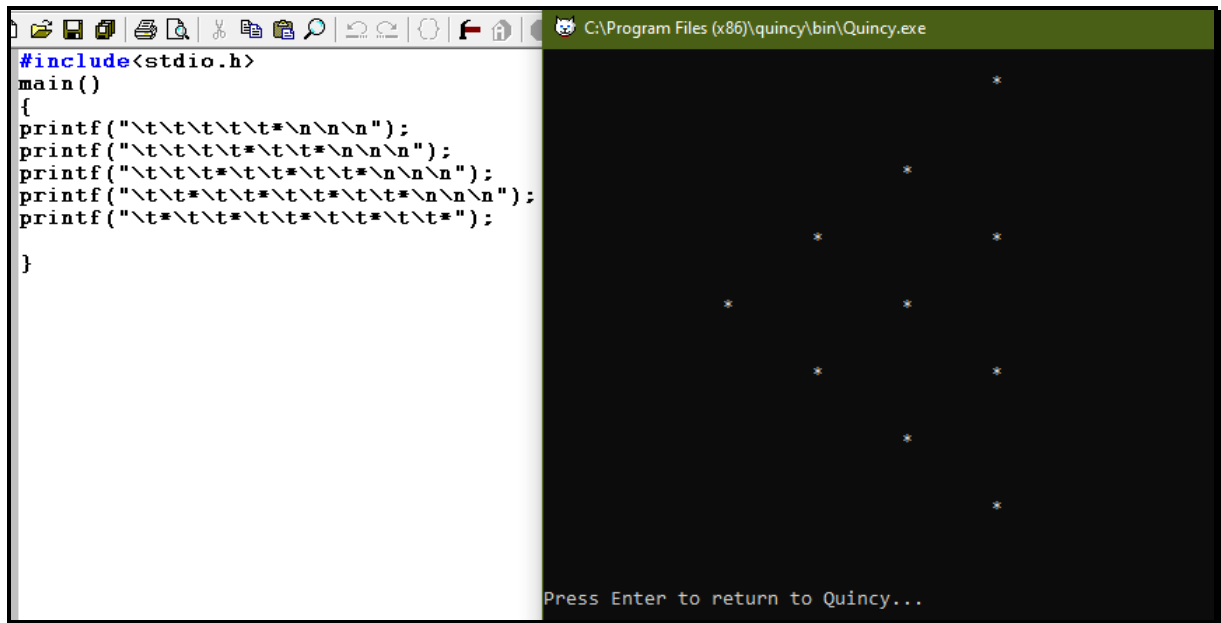
```
#include<stdio.h>  
main()  
{  
printf("*\n\n\n");  
printf("\t*\n\n\n");  
printf("*\t\t*\n\n\n");  
printf("\t*\t\t\t*\n\n\n");  
printf("*\t\t*\n\n\n");  
printf("\t*\n\n\n");  
printf("*\n\n\n\n\n");  
}
```

d. Write a C program to display the following output;

```
      *  
  
    *  
  
  *      *  
  
*      *  
  
  *      *  
  
    *  
  
      *
```

Light.

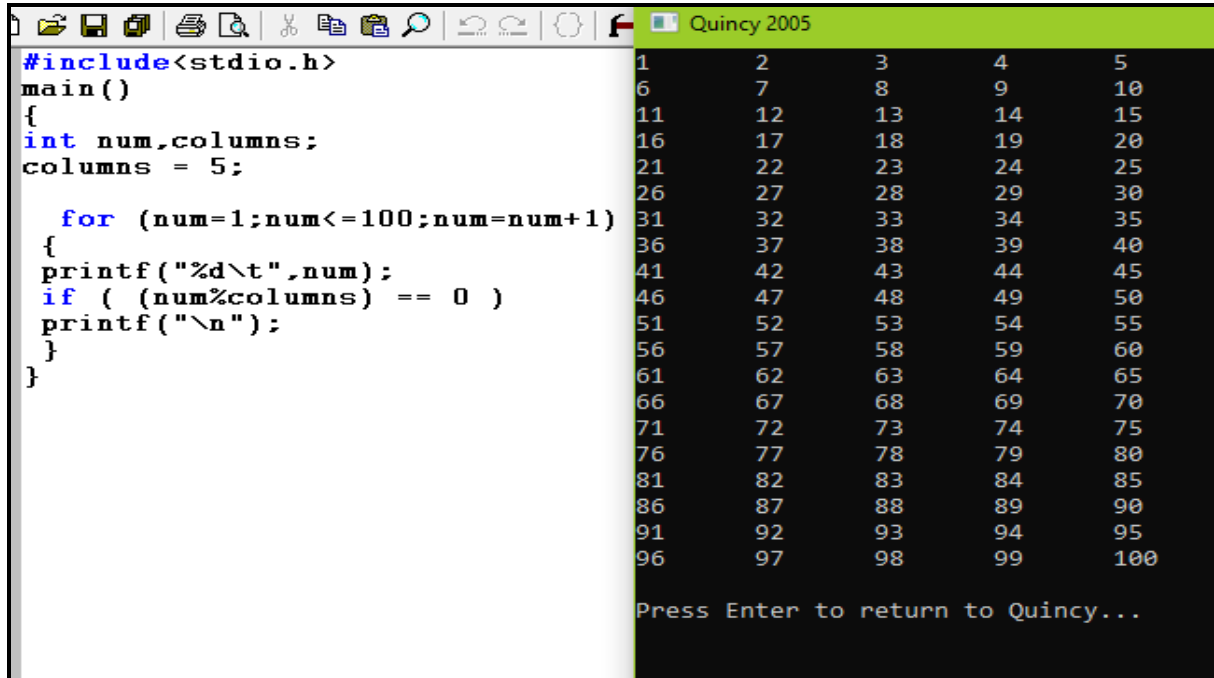
Light.



The screenshot shows a C program in the Quincy IDE. The code on the left uses `printf` with tab characters to print a star pattern on the right. The pattern consists of 10 stars arranged in a diamond shape: 1 star in the first row, 2 in the second, 3 in the third, 4 in the fourth, 5 in the fifth, 4 in the sixth, 3 in the seventh, 2 in the eighth, 1 in the ninth, and 1 in the tenth. The prompt "Press Enter to return to Quincy..." is visible at the bottom right.

```
#include<stdio.h>
main()
{
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
printf("\t\t\t\t\t*\n\n\n");
}
```

8. Write a C program that prints the numbers 1 to 100 using 5 columns. Have each number separated from the next by a tab.



The screenshot shows a C program in the Quincy 2005 IDE. The code on the left uses a `for` loop and `printf` to print numbers 1 to 100 in 5 columns, separated by tabs. The output on the right shows the numbers 1 through 100 arranged in 5 columns, with each number separated from the next by a tab. The prompt "Press Enter to return to Quincy..." is visible at the bottom right.

```
#include<stdio.h>
main()
{
int num,columns;
columns = 5;

for (num=1;num<=100;num=num+1)
{
printf("%d\t",num);
if ( (num%columns) == 0 )
printf("\n");
}
}
```

9. Write a C program to display the following output

Light.

Light.

```
* * * *  
* * *  
* *  
*
```

```
1  #include <stdio.h>  
2  main()  
3  {  
4  
5      printf("    * * * *\n");  
6      printf("    * * *\n");  
7      printf("    * * \n");  
8      printf("    *\n");  
9  }
```

10. Write a C program to find the eligibility of admission for a professional course based on the following criteria: Eligibility Criteria: Grade in Maths and Physics should be minimum of C.

```
#include <stdio.h>  
main ( )  
{  
int grades;  
printf ("Enter the Maths grades: \n");  
scanf ("%d",& grades);  
{  
if ( grades >= 70 )  
printf ( "A\n\n" );  
else if ( grades >= 60 )  
printf ( "B+\n\n");
```

Light.

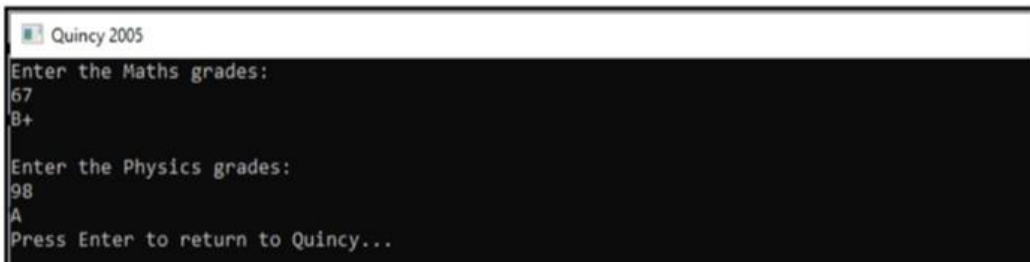
```
else if ( grades >= 50 )
    printf ("B\n\n");
else if ( grades >= 40 )
    printf ("C\n\n");
else if ( grades >= 0 )
    printf ("Admission is not eligible, since the grades haven't reached the
target\n\n");
}
{
printf ("Enter the Physics grades: \n");
scanf ("%d",& grades);
if ( grades >= 70 )

    printf ("A");
else if ( grades >= 60 )
    printf ("B+");
else if ( grades >= 50 )
    printf ("B");
else if ( grades >= 40 )

    printf ("C");
else if ( grades >= 0 )
    printf ("Admission is not eligible, since the grades haven't reached the target");
}
}
```

After running the program.

i.



```
Quincy 2005
Enter the Maths grades:
67
B+

Enter the Physics grades:
98
A
Press Enter to return to Quincy...
```


Light.

ii.

```
C:\Program Files (x86)\quincy\bin\Quincy.exe
Enter the Maths grades:
67
8+
Enter the Physics grades:
20
Admission is not eligible,since the grades haven't reached the target
Press Enter to return to Quincy...
```

iii.

```
Enter the Maths grades:
34
Admission is not eligible,since the grades haven't reached the target
Enter the Physics grades:
12
Admission is not eligible,since the grades haven't reached the target
Press Enter to return to Quincy...
```

11. Write a C program to read n numbers from keyboard and find their sum and Average.

```
#include<stdio.h>
main()
{
    int i,n,sum=0,average;
    printf("\n How many numbers to be executed ? : ");
    scanf("%d",&n);

    for(i=1; i<=n; i++)
    {
        sum=sum+i;
        average=sum/n;
    }
    printf(" The sum and average are: %d,%d \n",sum,average);
}
```

```
How many numbers to be executed ? : 4
The sum and average are: 10,2
Press Enter to return to Quincy...
```

```
#include<stdio.h>
main()
{
    int i,n,sum=0,average;
    printf("\n How many numbers to be executed ? : ");
    scanf("%d",&n);

    for(i=1; i<=n; i++)
    {
        sum=sum+i;
        average=sum/n;
    }
    printf(" The sum and average are: %d,%d \n",sum,average);
}
```

```
How many numbers to be executed ? : 5
The sum and average are: 15,3
Press Enter to return to Quincy...
```

Kuwen na maandalizi mema ya UE.

Light.